

# Measuring Muscle, Automatically

Geometry-aware deep learning for muscle architecture estimation in B-mode ultrasound

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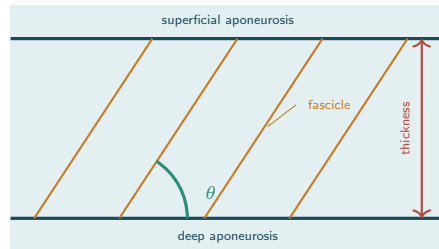
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# Motivation: the analysis bottleneck

- B-mode ultrasound is cheap, portable and non-ionising — the standard tool for imaging skeletal muscle.
- Three parameters describe *muscle architecture*: **muscle thickness**, **pennation angle**, **fascicle length**.
- These are still measured **by hand**: slow, subjective, and dependent on operator experience.
- Existing automated tools exist, but their authors report performance is **not yet sufficient for out-of-the-box use** [7].

## The three measurements



## Problem statement & data

**Task.** Given a B-mode ultrasound image or video frame of a lower-limb muscle, estimate:

1. **muscle thickness** (mm) — separation of the two aponeuroses;
2. **pennation angle** ( $^{\circ}$ ) — angle between fascicle and deep aponeurosis;
3. **fascicle length** (mm) — derived from thickness and orientation.

**Data.** Public musculoskeletal ultrasonography benchmarks indexed by UMUD [7]:

- *Multi-expert set*: 35 architecture images analysed independently by **six expert operators**;
- *Expert-drawn set*: 250 images with expert-traced fascicles (vastus lateralis, young + older adults);
- *Labelled training sets*: fascicle and aponeurosis annotations for supervised learning;
- *Wider repository*: 14 datasets,  $\sim 76,000$  images and  $\sim 2,700$  videos from  $\sim 1,900$  participants.

**Evaluation.** Absolute error per parameter against expert ground truth, referenced to **inter-expert variability**.

# Literature Review

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## Literature review I — Q1 journal papers

| Paper                                                                                          | Venue                                    | What we take from it                                                                                                         |
|------------------------------------------------------------------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Ritsche et al., <i>Fully automated analysis of muscle architecture with DL_Track_US</i> [1]    | Ultrasound in Medicine & Biology, 2024   | The <b>reference baseline</b> : U-Net segmentation of fascicles and aponeuroses; reports the residual gap on video sequences |
| Marzola et al., <i>Deep learning segmentation of transverse musculoskeletal ultrasound</i> [2] | Computers in Biology and Medicine, 2021  | Large-scale (3,917 images, 3 muscles) segmentation study; evidence on <b>cross-muscle generalisation</b>                     |
| Leitner et al., <i>Human-centred ML for muscle–tendon junction tracking</i> [3]                | IEEE Trans. Biomedical Engineering, 2022 | <b>Temporal tracking</b> of anatomical landmarks across devices and subjects; motivates our video module                     |

## Literature review II — A\*/A conference papers

| Paper                                                              | Venue           | What we take from it                                                                                                     |
|--------------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------|
| Ronneberger, Fischer, Brox, <i>U-Net</i> [4]                       | MICCAI 2015 (A) | The encoder–decoder architecture underlying every baseline in this field; our segmentation backbone                      |
| Kirillov et al., <i>Segment Anything</i> [5]                       | ICCV 2023 (A*)  | Promptable <b>foundation-model segmentation</b> ; tests whether large-scale pretraining transfers to speckled ultrasound |
| Teed & Deng, <i>RAFT: Recurrent All-Pairs Field Transforms</i> [6] | ECCV 2020 (A*)  | Dense <b>optical flow</b> via iterative refinement; basis for propagating fascicle geometry across frames                |

# Synthesis: the gap our project targets

## What prior work does

- Segments muscle structures well on *single images*
- Derives geometry as a *post-processing* step outside the learned model
- Evaluates against *one* expert annotator
- Trains and tests largely on *one* muscle and device

⇒ Couple segmentation with explicit orientation estimation and temporal filtering.

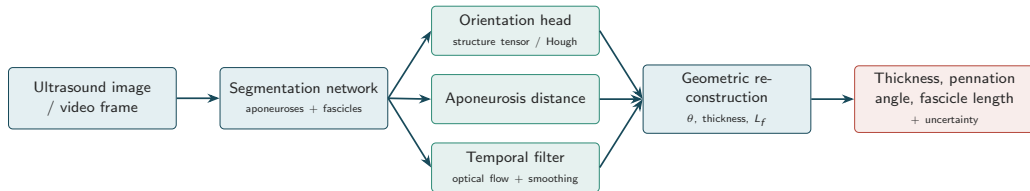
## What is missing

- **Geometry-aware** estimation: orientation treated as a first-class output
- **Temporal coherence** across video frames
- **Uncertainty** calibrated against multi-expert disagreement
- **Generalisation** across muscles and devices

## Proposed Approach

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# Pipeline



Segmentation supplies structure; the orientation head supplies direction; the temporal filter enforces frame-to-frame consistency.

## Evaluation

- Mean absolute error per parameter (mm, mm, degrees)
- Error compared against **inter-expert spread** — is the model within human disagreement?
- Segmentation quality: IoU / Dice on aponeurosis and fascicle masks
- Breakdown by **muscle** and **device** to expose generalisation gaps
- Ablations: remove orientation head; remove temporal filter

## Expected contributions

1. A **geometry-aware** architecture estimator that predicts fascicle orientation explicitly rather than recovering it by post-processing.
2. A **temporally consistent** extension for video sequences, addressing the weakest point of current automated tools.
3. An error analysis referenced to **multi-expert disagreement**, and a generalisation study **across muscles and devices**.

### One-sentence summary

We turn a manual, operator-dependent measurement into a reproducible vision pipeline — and report honestly how close it gets to expert agreement.

- [1] P. Ritsche, M. V. Franchi, O. Faude, T. Finni, O. Seynnes, N. J. Cronin. Fully automated analysis of muscle architecture from B-mode ultrasound images with DL\_Track\_US. *Ultrasound in Medicine & Biology*, 50(2):258–267, 2024.
- [2] F. Marzola, N. van Alfen, J. Doorduyn, K. M. Meiburger. Deep learning segmentation of transverse musculoskeletal ultrasound images for neuromuscular disease assessment. *Computers in Biology and Medicine*, 135:104623, 2021.
- [3] C. Leitner, R. Jarolim, B. Englmaier, A. Kruse, K. A. L. Hernandez, A. Konrad, E. Y.-S. Su, J. Schröttner, L. A. Kelly, G. A. Lichtwark, M. Tilp, C. Baumgartner. A human-centered machine-learning approach for muscle–tendon junction tracking in ultrasound images. *IEEE Transactions on Biomedical Engineering*, 69(6):1920–1930, 2022.
- [4] O. Ronneberger, P. Fischer, T. Brox. U-Net: Convolutional networks for biomedical image segmentation. *MICCAI*, pp. 234–241, 2015.

- [5] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, P. Dollár, R. Girshick. Segment Anything. *ICCV*, pp. 4015–4026, 2023.
- [6] Z. Teed, J. Deng. RAFT: Recurrent all-pairs field transforms for optical flow. *ECCV*, pp. 402–419, 2020.
- [7] P. Ritsche, F. Sarto, F. Santini, C. Leitner, M. V. Franchi, O. Faude, T. Finni, O. Seynnes, N. Cronin. UMUD: a web application for easy access to musculoskeletal ultrasonography datasets. *BMC Medical Imaging*, 26:139, 2026.